JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2026 28th January Morning Shift

Question Paper with Answer Key

75	ENGLISH	DARK
QUESTIONS	LANGUAGE	FORMAT

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : The number of pairs, from the following, in which both the ions are coloured in aqueous solution is 3.

$[Sc^{3+}, Ti^{3+}]$, $[Mn^{2+}, Cr^{2+}]$, $[Cu^{2+}, Zn^{2+}]$ and $[Ni^{2+}, Ti^{4+}]$

Statement II : Th^{4+} is the strongest reducing agent among Th^{4+}, Ce^{4+}, Gd^{3+} and Eu^{2+}.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Q2. MCQ (Single Correct Answer)

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$20.0 dm^3$ of an ideal gas ' X ' at 600 K and 0.5 MPa undergoes isothermal reversible expansion until pressure of the gas is 0.2 MPa . Which of the following option is correct?

(Given: $\log 2 = 0.3010$ and $\log 5 = 0.6989$)

(A) $w = -9.1\,kJ, \Delta U = 0, \Delta H = 0, q = 9.1\,kJ$

(B) $w = 9.1\,J, \Delta U = 9.1\,J, \Delta H = 0; q = 0$

(C) $w = -3.9\,kJ, \Delta U = 0, \Delta H = 0; q = 3.9\,kJ$

(D) $w = +4.1\,kJ, \Delta U = 0, \Delta H = 0; q = -4.1\,kJ$

Q3. MCQ (Single Correct Answer)

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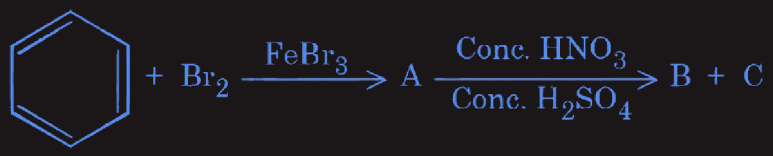
In period 4 of the periodic table, the elements with highest and lowest atomic radii are respectively.

(A) Rb & Br (B) Na & Cl (C) K&Br (D) K&Se

Q4. MCQ (Single Correct Answer)

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Method used for separation of mixture of products (B and C) obtained in the following reaction is



(A) simple distillation

(B) sublimation

(C) fractional distillation

(D) steam distillation

Q5. MCQ (Single Correct Answer)

Regarding the hydrides of group 15 elements $EH_3(E = N, P, As, Sb)$, select the correct statement from the following :

A. The stability of hydrides decreases down the group.

B. The basicity of hydrides decreases down the group.

C. The reducing character increases down the group.

D. The boiling point increases down the group.

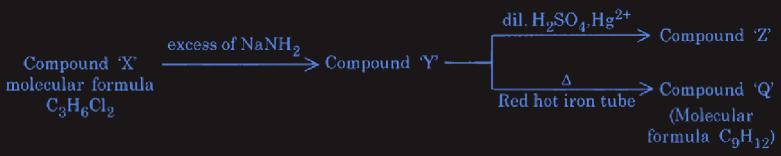
Choose the correct answer from the options given below :

(A) A, B & C only (B) A, B, C& D (C) A & D only (D) B & C only

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Q6. MCQ (Single Correct Answer)

Given below are two statements for the following reaction sequence.



Statement I : Compound ' Z ' will give yellow precipitate with NaOI .

Statement II : Compound ' Q ' has two different types of ' H ' atoms (aromatic : aliphatic) in the ratio $1:3$.

In the light of the above statements, choose the ***correct*** answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

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Q7. MCQ (Single Correct Answer)

$Ph - CH = CH_2 \xrightarrow[HBr]{(PhCOO)_2}$ Product

Consider the above reaction

A. The reaction proceeds through a more stable radical intermediate.

B. The role of peroxide is to generate $H^{\bullet}$ (Hydrogen radical).

C. During this reaction, benzene is formed as a byproduct.

D. 1-Bromo-2-phenylethane is formed as the minor product.

E. The same reaction in absence of peroxide proceeds via carbocation intermediate.

Identify the correct statements. Choose the correct answer from the options given below :

(A) A & E Only (B) A, B & D Only (C) A, C & E Only (D) C, D & E Only

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Q8. MCQ (Single Correct Answer)

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An organic compound undergoes first order decomposition. The time taken for decomposition to $\left(\frac{1}{8}\right)^{\text{th}}$ and $\left(\frac{1}{10}\right)^{\text{th}}$ of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively.

What is the value of $\frac{t_{1/8}}{t_{1/10}} \times 10$?

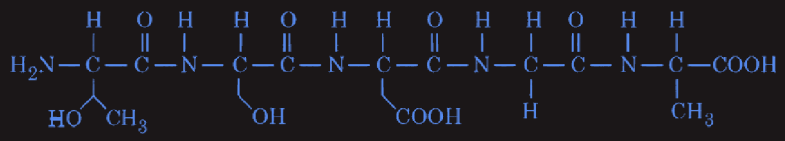
$(\log 2 = 0.3)$

(A) 30 (B) 9 (C) 3 (D) 0.9

Q9. MCQ (Single Correct Answer)

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In the given pentapeptide, find out an essential amino acid (Y) and the sequence present in the pentapeptide :



Choose the correct answer from the options given below :

(A) (Y) Serine (Sequence) Thr-Ser-Asp-Ala-Gly

(B) (Y) Serine (Sequence) Ser-Asp-Thr-Ala-Gly

(C) (Y) Threonine (Sequence) Ser-Thr-Asp-Gly-Ala

(D) (Y) Threonine (Sequence) Thr-Ser-Asp-Gly-Ala

Q10. MCQ (Single Correct Answer)

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At $T(K), 2$ moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of ideal solution formed is 320 mm Hg . At this stage, one mole of A and one mole of B are added to the solution. The vapour pressure is now measured as 328.6 mm Hg . The vapour pressure (in mm Hg) of A and B are respectively:

(A) 400, 300 (B) 600, 400 (C) 500, 200 (D) 300, 200

Q11. MCQ (Single Correct Answer)

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The wave numbers of three spectral lines of H atom are considered. Identify the set of spectral lines belonging to Balmer series.

(R = Rydberg constant)

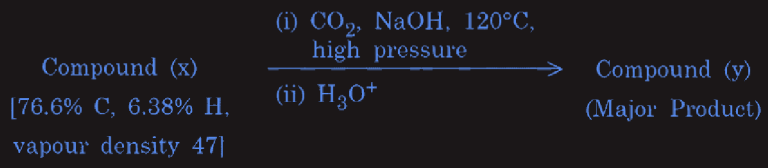
(A) $\frac{5R}{36}, \frac{3R}{16}, \frac{21R}{100}$ (B) $\frac{7R}{144}, \frac{3R}{16}, \frac{16R}{255}$

(C) $\frac{3R}{4}, \frac{3R}{16}, \frac{7R}{144}$ (D) $\frac{5R}{36}, \frac{8R}{9}, \frac{15R}{16}$

Q12. MCQ (Single Correct Answer)

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Consider the following reaction sequence :



Compound (y) develops characteristic colour with neutral $FeCl_3$ solution.

Identify the **INCORRECT** statement from the following for the above sequence.

(A) Compound y will dissolve in $NaHCO_3$ and evolve a gas.

(B) Compound x is more acidic than compound y .

(C) Both compounds x and y will dissolve in NaOH .

(D) Both compounds x and y will burn with sooty flame.

Q13. MCQ (Single Correct Answer)

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CORRECT order of stability for the following is

$CH_2 = CH^-, CH_3 - CH_2^-, CH \equiv C^-$

(A) $CH \equiv C^- > CH_3 - CH_2^- > CH_2 = CH^-$

(B) $CH_2 = CH^- > CH \equiv C^- > CH_3 - CH_2^-$

(C) $CH \equiv C^- > CH_2 = CH^- > CH_3 - CH_2^-$

(D) $CH_3 - CH_2^- > CH_2 = CH^- > CH \equiv C^-$

Q14. MCQ (Single Correct Answer)

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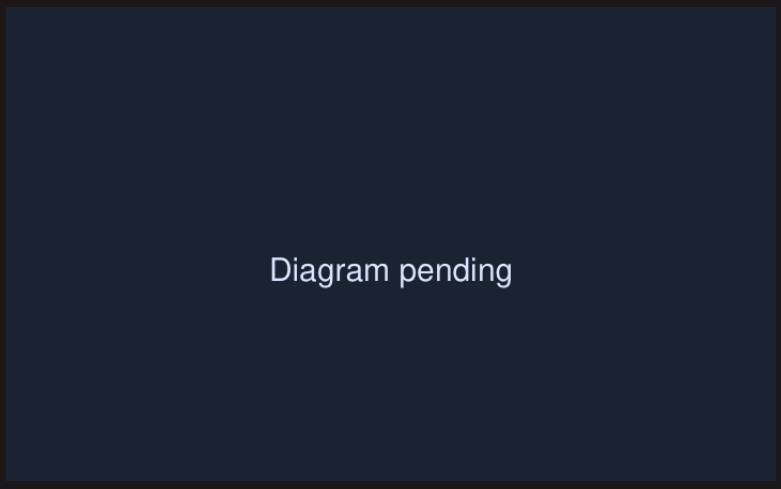


Figure 1. electron probability density for 2 s orbital

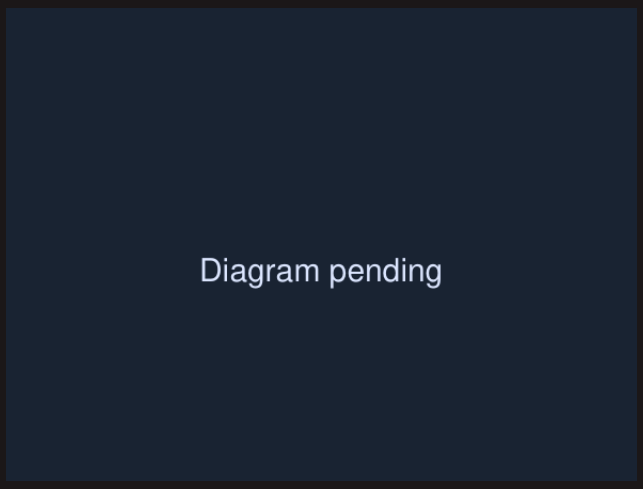


Figure 2. wave function for 2s orbital

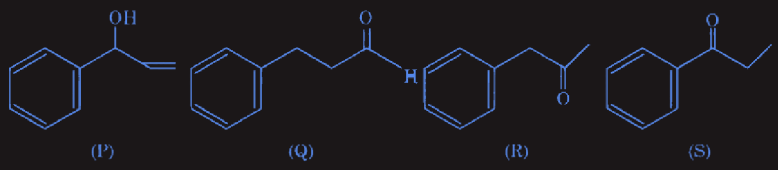
Which of the following point in Figure 2 most accurately represents the nodal surface as shown in Figure 1?

(A) D (B) B (C) C (D) A

Q15. MCQ (Single Correct Answer)

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Given below are the four isomeric compounds (P, Q, R, S)



Identify correct statements from below.

A. Q, R and S will give precipitate with 2, 4-DNP.

B. P and Q will give positive Baeyer's test.

C. Q and R will give sooty flame.

D. R and S will give yellow precipitate with $\mathrm{I_2/NaOH}$.

E. Q alone will deposit silver with Tollens' reagent

Choose the correct option.

(A) C and E only

(B) A and E only

(C) A, B, D and E only

(D) A, C and E only

Q16. MCQ (Single Correct Answer)

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Consider a weak base ' B ' of $\mathrm{pK_b} = 5.699$. ' x ' mL of 0.02 M HCl and ' y ' mL of 0.02 M weak base ' B ' are mixed to make 100 mL of a buffer of pH 9 at $25^{\circ}\mathrm{C}$. The values of ' x ' and ' y ' respectively are :

[Given : $\log 2 = 0.3010, \log 3 = 0.4771, \log 5 = 0.699$]

(A)

x	y
42.7	57.3

(B)

x	y
14.3	85.7

(C)

x	y
85.7	14.3

(D)

x	y
11.1	88.9

Q17. MCQ (Single Correct Answer)

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Given below are two statements:

Statement I : Griess-Ilosvay test is used for the detection of nitrite ion, which involves the use of sulphanilic acid and α-naphthylamine reagent.

Statement II : In the above test, sulphanilic acid is diazotized by the acidified nitrite ion, which on further coupling with α-naphthylamine forms an azo-dye.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are false

(C) Statement I is false but Statement II is true

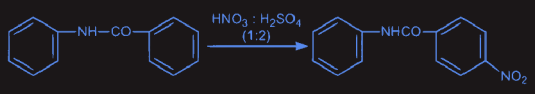
(D) Both Statement I and Statement II are true

Q18. MCQ (Single Correct Answer)

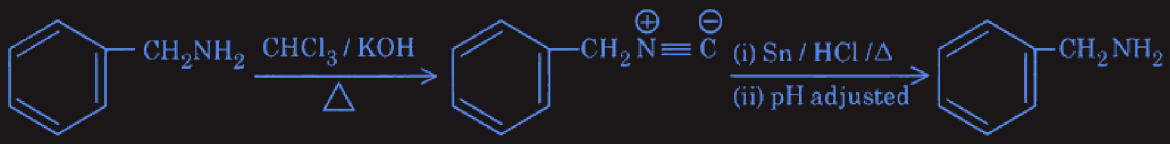
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Consider the following reactions giving major product. Identify the correct reaction.

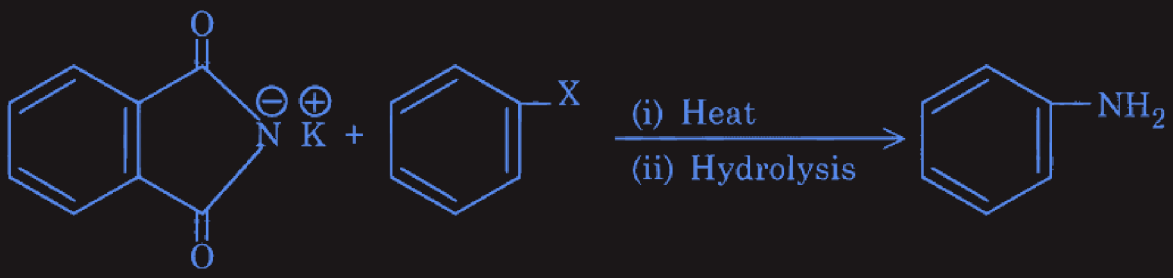
(A)



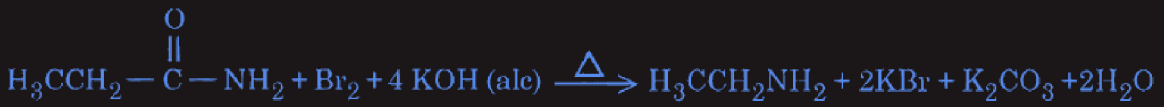
(B)



(C)



(D)



Q19. MCQ (Single Correct Answer)

The correct statement among the following is:

(A) $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ are diamagnetic and $Ni(CO)_4$ is paramagnetic.

(B) $Ni(CO)_4$ is diamagnetic and $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ are paramagnetic.

(C) $Ni(CO)_4$ and $[NiCl_4]^{2-}$ are diamagnetic and $[Ni(CN)_4]^{2-}$ is paramagnetic.

(D) $Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$ are diamagnetic and $[NiCl_4]^{2-}$ is paramagnetic.

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Q20. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : The number of species among BF_4^-, SiF_4, XeF_4 and SF_4, that have unequal $E-F$ bond lengths is two. Here, E is the central atom.

Statement II : Among O_2^-, O_2^{2-}, F_2 and O_2^+, O_2^- has the highest bond order.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are true

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Q21. Numerical

Consider the following redox reaction taking place in acidic medium

$BH_4^-(aq) + ClO_3^-(aq) \longrightarrow H_2BO_3^-(aq) + Cl^-(aq)$

If the Nernst equation for the above balanced reaction is

$E_{cell} = E^\circ_{cell} - \frac{RT}{nF}\ln Q$,

then the value of n is ____ .(Nearest integer)

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Q22. Numerical

X is the number of geometrical isomers exhibited by $[Pt(NH_3)(H_2O)BrCl]$.

Y is the number of optically inactive isomer(s) exhibited by $[CrCl_2(ox)_2]^{3-}$

Z is the number of geometrical isomers exhibited by $[Co(NH_3)_3(NO_2)_3]$.

The value of $X + Y + Z$ is ____ .

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Q23. Numerical

Consider the dissociation equilibrium of the following weak acid

$HA \rightleftharpoons H^+(aq) + A^-(aq)$

If the pKa of the acid is 4 , then the pH of 10 mM HA solution is ____ .(Nearest integer)

[Given: The degree of dissociation can be neglected with respect to unity]

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Q24. Numerical

0.53 g of an organic compound (x) when heated with excess of nitric acid (concentrated) and then with silver nitrate gave 0.75 g of silver bromide precipitate. 1.0 g of (x) gave 1.32 g of $\mathrm{CO_2}$ gas on combustion. The percentage of hydrogen in the compound (x) is ____ %. [Nearest Integer]

[Given: Molar mass in $\mathrm{gmol^{-1}H}: 1, \mathrm{C}: 12, \mathrm{Br}: 80, \mathrm{Ag}: 108, \mathrm{O}: 16$; Compound (x) : $\mathrm{C_xH_yBr_z}$]

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Q25. Numerical

500 mL of 1.2 M KI solution is mixed with 500 mL of $0.2\mathrm{MKMnO_4}$ solution in basic medium. The liberated iodine was titrated with standard $0.1\mathrm{MNa_2\,S_2O_3}$ solution in the presence of starch indicator till the blue color disappeared. The volume (in L) of $\mathrm{Na_2\,S_2O_3}$ consumed is ____ . (Nearest integer)

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

The area of the region $\mathrm{R} = \{(x, y) : xy \leq 8, 1 \leq y \leq x^2, x \geq 0\}$ is

(A) $\frac{2}{3}(20\log_e(2) + 9)$

(B) $\frac{1}{3}(40\log_e(2) + 27)$

(C) $\frac{1}{3}(49\log_e(2) - 15)$

(D) $\frac{2}{3}(24\log_e(2) - 7)$

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Q27. MCQ (Single Correct Answer)

If $g(x) = 3x^2 + 2x - 3, f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to:

(A) $\frac{7}{2}$

(B) $-\frac{25}{6}$

(C) $\frac{25}{6}$

(D) $-\frac{7}{2}$

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Q28. MCQ (Single Correct Answer)

A bag contains 10 balls out of which k are red and $(10 - k)$ are black, where $0 \leq k \leq 10$. If three balls are drawn at random without replacement and all of them are found to be black, then the probability that the bag contains 1 red and 9 black balls is:

(A) $\frac{7}{110}$

(B) $\frac{7}{11}$

(C) $\frac{7}{55}$

(D) $\frac{14}{55}$

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Q29. MCQ (Single Correct Answer)

The mean and variance of 10 observations are 9 and 34.2 , respectively. If 8 of these observations are $2, 3, 5, 10, 11, 13, 15, 21$, then the mean deviation about the median of all the 10 observations is

(A) 7

(B) 4

(C) 5

(D) 6

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Q30. MCQ (Single Correct Answer)

If α, β, where $\alpha < \beta$, are the roots of the equation $\lambda x^2 - (\lambda + 3)x + 3 = 0$ such that $\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{3}$, then the sum of all possible values of λ is

(A) 2 (B) 6 (C) 8 (D) 4

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Q31. MCQ (Single Correct Answer)

Let $y = x$ be the equation of a chord of the circle C_1 (in the closed half-plane $x \geq 0$) of diameter 10 passing through the origin. Let C_2 be another circle described on the given chord as its diameter. If the equation of the chord of the circle C_2, which passes through the point $(2, 3)$ and is farthest from the center of C_2, is $x + ay + b = 0$, then $a - b$ is equal to

(A) -6 (B) 10 (C) 6 (D) -2

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Q32. MCQ (Single Correct Answer)

Let $S = \{x^3 + ax^2 + bx + c : a, b, c \in N$ and $a, b, c \leq 20\}$ be a set of polynomials. Then the number of polynomials in S , which are divisible by $x^2 + 2$, is

(A) 6 (B) 120 (C) 20 (D) 10

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Q33. MCQ (Single Correct Answer)

Let A, B and C be three 2×2 matrices with real entries such that $B = (I + A)^{-1}$ and $A + C = I$.

If $BC = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$ and $CB \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$, then $x_1 + x_2$ is

(A) 4 (B) 2 (C) 0 (D) -2

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Q34. MCQ (Single Correct Answer)

Let ABC be an equilateral triangle with orthocenter at the origin and the side BC on the line $x + 2\sqrt{2}y = 4$. If the co-ordinates of the vertex A are (α, β), then the greatest integer less than or equal to $|\alpha + \sqrt{2}\beta|$ is

(A) 5 (B) 4 (C) 2 (D) 3

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Q35. MCQ (Single Correct Answer)

Let f be a polynomial function such that $f\left(x^2 + 1\right) = x^4 + 5x^2 + 2$, for all $x \in \mathbb{R}$.

Then $\int_0^3 f(x)dx$ is equal to

(A) $\frac{33}{2}$ (B) $\frac{5}{3}$ (C) $\frac{27}{2}$ (D) $\frac{41}{3}$

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Q36. MCQ (Single Correct Answer)

If $\int\left(\frac{1-5\cos^2 x}{\sin^5 x\cos^2 x}\right)dx = f(x)+C$, where C is the constant of integration, then $f\left(\frac{\pi}{6}\right)-f\left(\frac{\pi}{4}\right)$ is equal to

(A) $\frac{1}{\sqrt{3}}(26-\sqrt{3})$ (B) $\frac{4}{\sqrt{3}}(8-\sqrt{6})$ (C) $\frac{1}{\sqrt{3}}(26+\sqrt{3})$ (D) $\frac{2}{\sqrt{3}}(4+\sqrt{6})$

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Q37. MCQ (Single Correct Answer)

Let z be a complex number such that $|z-6|=5$ and $|z+2-6i|=5$. Then the value of $z^3+3z^2-15z+141$ is equal to :

(A) 61 (B) 37 (C) 42 (D) 50

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Q38. MCQ (Single Correct Answer)

The value of $\sum_{k=1}^{\infty}(-1)^{k+1}\left(\frac{k(k+1)}{k!}\right)$ is

(A) e/2 (B) $\sqrt{e}$ (C) 2/e (D) 1/e

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Q39. MCQ (Single Correct Answer)

The common difference of the A.P.: $a_1, a_2, \ldots, a_m$ is 13 more than the common difference of the A.P.: $b_1, b_2, \ldots, b_n$. If $b_{31}=-277$, $b_{43}=-385$ and $a_{78}=327$, then a_1 is equal to

(A) 21 (B) 19 (C) 24 (D) 16

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Q40. MCQ (Single Correct Answer)

The value of

$$\lim_{x\to 0}\frac{\log_e\left(\sec(ex)\cdot\sec\left(e^2x\right)\cdot\ldots\cdot\sec\left(e^{10}x\right)\right)}{e^2-e^{2\cos x}}$$

is equal to

(A) $\frac{\left(e^{10}-1\right)}{2e^2\left(e^2-1\right)}$ (B) $\frac{\left(e^{20}-1\right)}{2e^2\left(e^2-1\right)}$

(C) $\frac{\left(e^{10}-1\right)}{2\left(e^2-1\right)}$ (D) $\frac{\left(e^{20}-1\right)}{2\left(e^2-1\right)}$

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Q41. MCQ (Single Correct Answer)

Let $S=\{1,2,3,4,5,6,7,8,9\}$. Let x be the number of 9-digit numbers formed using the digits of the set S such that only one digit is repeated and it is repeated exactly twice. Let y be the number of 9 -digit numbers formed using the digits of the set S such that only two digits are repeated and each of these is repeated exactly twice. Then,

(A) $56x=9y$ (B) $21x=4y$ (C) $45x=7y$ (D) $29x=5y$

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Q42. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution of the differential equation

$x\frac{dy}{dx} - \sin 2y = x^3\left(2 - x^3\right)\cos^2 y, x \neq 0.$

If $y(2) = 0$, then $\tan(y(1))$ is equal to

(A) $-\frac{7}{4}$ (B) $-\frac{3}{4}$ (C) $\frac{3}{4}$ (D) $\frac{7}{4}$

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Q43. MCQ (Single Correct Answer)

If $\frac{\tan(\mathrm{A}-\mathrm{B})}{\tan \mathrm{A}} + \frac{\sin^2 \mathrm{C}}{\sin^2 \mathrm{A}} = 1,\ \mathrm{A},\ \mathrm{B}, \mathrm{C} \in \left(0, \frac{\pi}{2}\right)$, then

(A) $\tan \mathrm{A}, \tan \mathrm{C}, \tan \mathrm{B}$ are in A.P.

(B) $\tan \mathrm{A}, \tan \mathrm{C}, \tan \mathrm{B}$ are in G.P.

(C) $\tan A, \tan B, \tan C$ are in G.P.

(D) $\tan A, \tan B, \tan C$ are in A.P.

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Q44. MCQ (Single Correct Answer)

If the distances of the point $(1, 2, a)$ from the line $\frac{x-1}{1} = \frac{y}{2} = \frac{z-1}{1}$ along the lines $\mathrm{L}_1 : \frac{x-1}{3} = \frac{y-2}{4} = \frac{z-a}{b}$ and $\mathrm{L}_2 : \frac{x-1}{1} = \frac{y-2}{4} = \frac{z-a}{c}$ are equal, then $a + b + c$ is equal to

(A) 4 (B) 6 (C) 7 (D) 5

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Q45. MCQ (Single Correct Answer)

For three unit vectors $\vec{a}, \vec{b}, \vec{c}$ satisfying

$|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$ and $|2\vec{a} + k\vec{b} + k\vec{c}| = 3$,

the positive value of k is :

(A) 4 (B) 5 (C) 6 (D) 3

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Q46. Numerical

For some $\theta \in \left(0, \frac{\pi}{2}\right)$, let the eccentricity and the length of the latus rectum of the hyperbola $x^2 - y^2\sec^2\theta = 8$ be e_1 and l_1, respectively, and let the eccentricity and the length of the latus rectum of the ellipse $x^2\sec^2\theta + y^2 = 6$ be e_2 and l_2, respectively. If $e_1^2 = e_2^2\left(\sec^2\theta + 1\right)$, then $\left(\frac{l_1 l_2}{e_1 e_2}\right)\tan^2\theta$ is equal to

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Q47. Numerical

In a G.P., if the product of the first three terms is 27 and the set of all possible values for the sum of its first three terms is $\mathbb{R} - (a, b)$, then $a^2 + b^2$ is equal to

____ .

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Q48. Numerical

If $k = \tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}\left(\frac{2}{3}\right)\right) + \tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2}{3}\right)\right)$, then

the number of solutions of the equation $\sin^{-1}(kx - 1) = \sin^{-1}x - \cos^{-1}x$ is ____.

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Q49. Numerical

The value of $\sum_{r=1}^{20}\left(\left|\sqrt{\pi\left(\int_0^r x|\sin \pi x|dx\right)}\right|\right)$ is ____

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Q50. Numerical

Let PQR be a triangle such that $\overrightarrow{PQ}=-2\hat{i}-\hat{j}+2\hat{k}$ and $\overrightarrow{\mathrm{PR}}=a\hat{\mathrm{i}}+b\hat{\mathrm{j}}-4\hat{\mathrm{k}}, a, b \in \mathrm{Z}$. Let S be the point on QR , which is equidistant from the lines PQ and PR . If $|\overrightarrow{\mathrm{PR}}|=9$ and $\overrightarrow{\mathrm{PS}}=\hat{\mathrm{i}}-7\hat{\mathrm{j}}+2\hat{\mathrm{k}}$, then the value of $3a-4b$ is ____

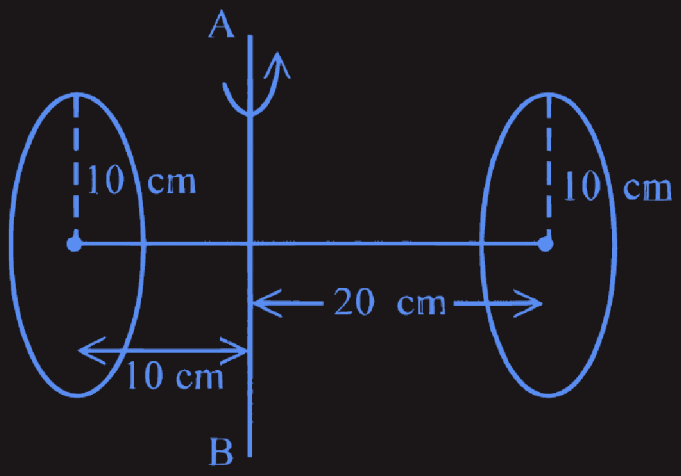
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Physics 25 questions

Q51. MCQ (Single Correct Answer)

Two circular discs of radius each 10 cm are joined at their centres by a rod of length 30 cm and mass 600 gm as shown in figure.

If the mass of each disc is 600 gm and applied torque between two discs is 43×10^5 dyne.cm, the angular acceleration of the discs about the given axis AB is ____ $\mathrm{rad/s^2}$.



(A) 100 (B) 22 (C) 27 (D) 11

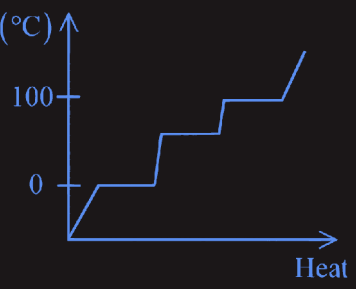
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Q52. MCQ (Single Correct Answer)

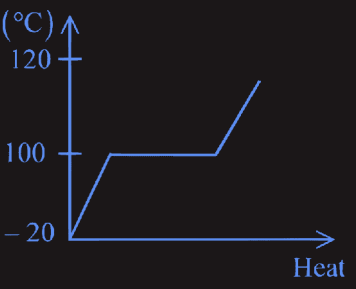
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Which of the following best represents the temperature versus heat supplied graph for water, in the range of $-20^{\circ}\mathrm{C}$ to $120^{\circ}\mathrm{C}$?

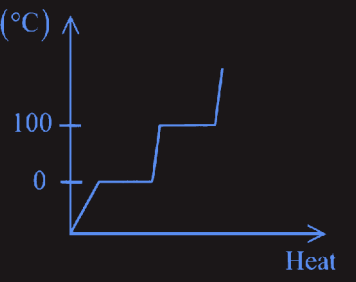
(A)



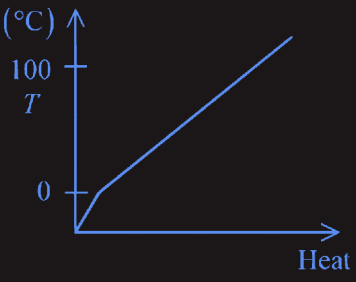
(B)



(C)



(D)

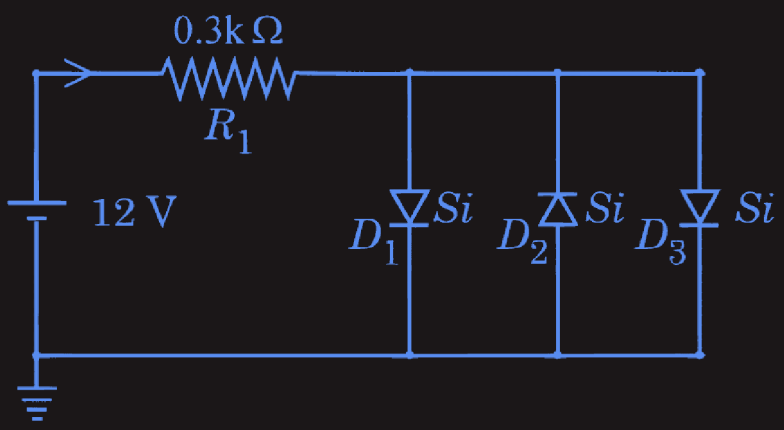


Q53. MCQ (Single Correct Answer)

Assuming in forward bias condition there is a voltage drop of 0.7 V across a silicon diode, the current through diode D_1 in the circuit is ____ mA .

(Assume all diodes in the given circuit are identical)

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(A) 17.6 (B) 18.8 (C) 20.15 (D) 11.7

Q54. MCQ (Single Correct Answer)

The electric current in the circuit is given as $i = i_o(t/T)$. The r.m.s current for the period $t = 0$ to $t = T$ is ____ .

(A) $\frac{i_o}{\sqrt{2}}$ (B) $\frac{i_o}{\sqrt{3}}$ (C) i_o (D) $\frac{i_o}{\sqrt{6}}$

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Q55. MCQ (Single Correct Answer)

A block of mass 5 kg is moving on an inclined plane which makes an angle of 30° with the horizontal. Friction coefficient between the block and inclined plane surface is $\frac{\sqrt{3}}{2}$. The force to be applied on the block so that the block will move down without acceleration is ____ N.

$(g = 10\ \mathrm{m/s^2})$.

(A) 25 (B) 12.5 (C) 15 (D) 7.5

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Q56. MCQ (Single Correct Answer)

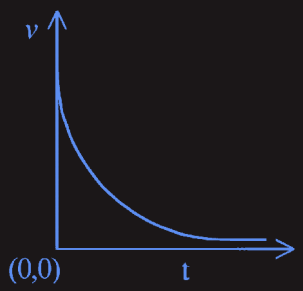
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A particle of mass m falls from rest through a resistive medium having resistive force, $F = -kv$, where v is the velocity of the particle and k is a constant. Which of the following graphs represents velocity (v) versus time (t) ?

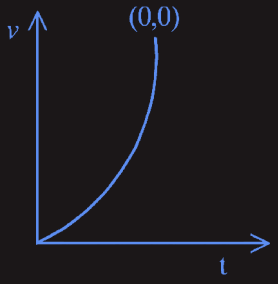
(A)

Diagram pending

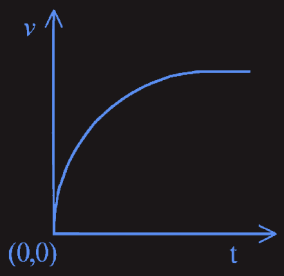
(B)



(C)



(D)



Q57. MCQ (Single Correct Answer)

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An atom ${}^{8}_{3}X$ is bombarded by shower of fundamental particles and in 10 s this atom absorbed 10 electrons, 10 protons and 9 neutrons. The percentage growth in the surface area of the nucleons is recorded by :

(A) 150% (B) 900% (C) 125% (D) 225%

Q58. MCQ (Single Correct Answer)

When both jaws of vernier callipers touch each other, zero mark of the vernier scale is right to zero mark of main scale, 4th mark on vernier scale coincides with certain mark on the main scale. While measuring the length of a cylinder, observer observes 15 divisions on main scale and 5th division of vernier scale coincides with a main scale division. Measured length of cylinder is ____ mm.

(Least count of Vernier calliper $= 0.1\ \text{mm}$)

(A) 15.5 (B) 15.4 (C) 15.9 (D) 15.1

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Q59. MCQ (Single Correct Answer)

The magnetic field at the centre of a current carrying circular loop of radius R is $16\mu\ \text{T}$. The magnetic field at a distance $x = \sqrt{3}R$ on its axis from the centre is

____ μT.

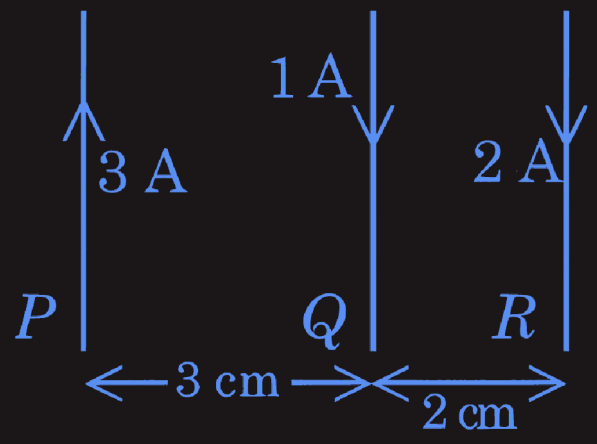
(A) 8 (B) 2 (C) 4 (D) $2\sqrt{2}$

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Q60. MCQ (Single Correct Answer)

Three long straight wires carrying current are arranged mutually parallel as shown in the figure. The force experienced by 15 cm length of wire Q is ____ .

$\left(\mu_o = 4\pi \times 10^{-7}\ \text{T.m/A}\right)$



(A) 6×10^{-6} N towards P (B) 6×10^{-6} N towards R

(C) 6×10^{-7} N towards P (D) 6×10^{-7} N towards R

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Q61. MCQ (Single Correct Answer)

The magnitudes of power of a biconvex lens (refractive index 1.5) and that of a plano-concave lens (refractive index $= 1.7$) are same. If the curvature of planoconcave lens exactly matches with the curvature of back surface of the biconvex lens, then ratio of radius of curvature of front and back surface of the biconvex lens is ____

(A) $2:5$ (B) $5:2$ (C) $12:5$ (D) $5:12$

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Q62. MCQ (Single Correct Answer)

The electric field of an electromagnetic wave travelling through a medium is given by $\vec{E}(x,t) = 25\sin\left(2.0 \times 10^{15}t - 10^{7}x\right)\hat{n}$ then the refractive index of the medium is ____ .

(All given measurement are in SI units)

(A) 2 (B) 1.2 (C) 1.5 (D) 1.7

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Q63. MCQ (Single Correct Answer)

Water drops fall from a tap on the floor, 5 m below, at regular intervals of time, the first drop strikes the floor when the sixth drop begins to fall. The height at which the fourth drop will be from ground, at the instant when the first drop strikes the ground is ____ m.

$\left(g = 10\ \mathrm{m/s^2}\right)$

(A) 3.8 (B) 4.0 (C) 4.2 (D) 2.5

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Q64. MCQ (Single Correct Answer)

For the two cells having same EMF E and internal resistance r, the current passing through the external resistor 6Ω is same when both the cells are connected either in parallel or in series. The value of internal resistance r is ____ Ω.

(A) 3 (B) 6 (C) 4 (D) 9

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Q65. MCQ (Single Correct Answer)

10 kg of ice at $-10^{\circ}\mathrm{C}$ is added to 100 kg of water to lower its temperature from 25 $^{\circ}\mathrm{C}$. Consider no heat exchange to surroundings. The decrement to the temperature of water is ____ $^{\circ}\mathrm{C}$.

(specific heat of ice $= 2100\ \mathrm{J/Kg.^{\circ}C}$, specific heat of water $= 4200\ \mathrm{J/Kg.^{\circ}C}$, latent heat of fusion of ice $= 3.36 \times 10^{5}\ \mathrm{J/Kg}$)

(A) 15 (B) 10 (C) 6.67 (D) 11.6

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Q66. MCQ (Single Correct Answer)

Two point charges of 1 nC and 2 nC are placed at the two corners of equilateral triangle of side 3 cm . The work done in bringing a charge of 3 nC from infinity to the third corner of the triangle is ____ $\mu\mathrm{J}$.

$\frac{1}{4\pi\epsilon_0} = 9 \times 10^{9}\ \mathrm{N \cdot\ m^2/C^2}$

(A) 3.3 (B) 27 (C) 2.7 (D) 5.4

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Q67. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : A plane wave after passing through prism remains as plane wave but passing through small pin hole may become spherical wave.

Statement II : The curvature of a spherical wave emerging from a slit will increase for increasing slit width.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q68. MCQ (Single Correct Answer)

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In the potentiometer, when the cell in the secondary circuit is shunted with 4Ω resistance, the balance is obtained at the length 120 cm of wire. Now when the same cell is shunted with 12Ω resistance, the balance is shifted to a length of 180 cm . The internal resistance of cell is ____ Ω

(A) 4 (B) 12 (C) 6 (D) 3

Q69. MCQ (Single Correct Answer)

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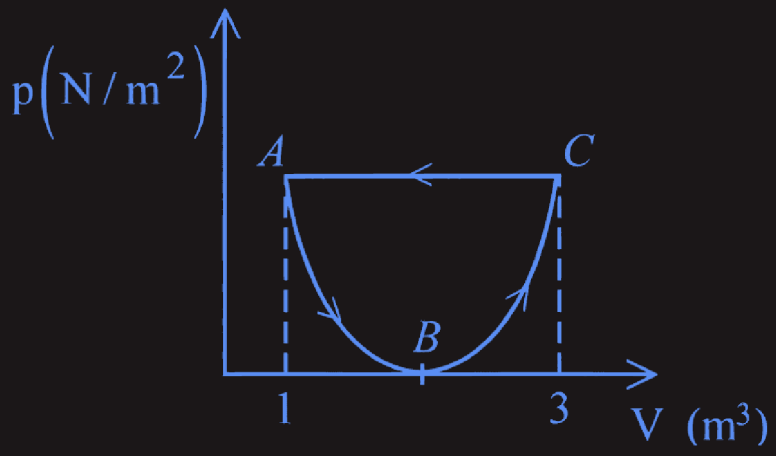
Two wires A and B made of different materials of lengths 6.0 cm and 5.4 cm , respectively and area of cross sections $3.0 \times 10^{-5}\ \mathrm{m}^2$ and $4.5 \times 10^{-5}\ \mathrm{m}^2$, respectively are stretched by the same magnitude under a given load. The ratio of the Young's modulus of A to that of B is $x : 3$. The value of x is ____

(A) 2 (B) 1 (C) 4 (D) 5

Q70. MCQ (Single Correct Answer)

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In the following $p - V$ diagram the equation of state along the curved path is given by $(V-2)^2 = 4ap$ where a is a constant. The total work done in the closed path is



(A) $+\frac{1}{3a}$ (B) $-\frac{1}{a}$ (C) $\frac{1}{2a}$ (D) $-\frac{1}{3a}$

Q71. Numerical

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A solid sphere of radius 10 cm is rotating about an axis which is at a distance 15 cm from its centre. The radius of gyration about this axis is $\sqrt{n}$ cm. The value of n is

Q72. Numerical

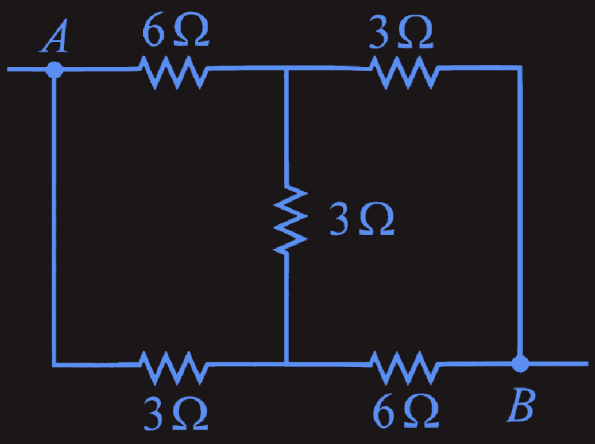
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The displacement of a particle, executing simple harmonic motion with time period T, is expressed as $x(t) = A\sin\omega t$, where A is the amplitude. The maximum value of potential energy of this oscillator is found at $t = T/2\beta$. The value of β is ____ .

Q73. Numerical

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The equivalent resistance between the points A and B in the following circuit is $\frac{x}{5}\Omega$. The value of x is ____ .



Q74. Numerical

A convex lens of refractive index 1.5 and focal length $f = 18\ \mathrm{cm}$ is immersed in water. The difference in focal lengths of the given lens when it is in water and in air is $\alpha \times \mathrm{f}$. The value of α is ____ .

(refractive index of water $= 4/3$)

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Q75. Numerical

The ratio of de Broglie wavelength of a deuteron with kinetic energy E to that of an alpha particle with kinetic energy $2E$, is $n : 1$. The value of n is ____ .

(Assume mass of proton $=$ mass of neutron)

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Answer Key

Chemistry

1. **B**	2. **A**	3. **C**	4. **C**	5. **A**
6. **B**	7. **C**	8. **B**	9. **D**	10. **C**
11. **A**	12. **B**	13. **C**	14. **B**	15. **D**
16. **B**	17. **D**	18. **D**	19. **D**	20. **A**
21. **24**	22. **6**	23. **3**	24. **4**	25. **3**

Mathematics

26. **D**	27. **A**	28. **D**	29. **C**	30. **B**
31. **D**	32. **D**	33. **C**	34. **B**	35. **A**
36. **B**	37. **D**	38. **D**	39. **B**	40. **D**
41. **B**	42. **D**	43. **B**	44. **C**	45. **B**
46. **8**	47. **90**	48. **1**	49. **210**	50. **37**

Physics

51. **D**	52. **C**	53. **B**	54. **B**	55. **B**
56. **D**	57. **C**	58. **D**	59. **B**	60. **B**
61. **B**	62. **C**	63. **C**	64. **B**	65. **B**
66. **C**	67. **C**	68. **A**	69. **D**	70. **D**
71. **265**	72. **2**	73. **21**	74. **3**	75. **2**

For numerical answers given as a range, any value within the range is correct.

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